

July 1, 2026

Professor Christa H. S. Bouwman
Managing Editor, *Journal of Banking and Finance*
Mays Business School, Texas A&M University

Dear Professor Bouwman,

I am pleased to submit “Leverage Direction Matters: Cross-Asset Evidence on GARCH Model Selection and Volatility Targeting” for consideration in the *Journal of Banking and Finance*. This paper examines how the direction of the asymmetric volatility response—the sign of the GJR-GARCH γ parameter—varies systematically across asset classes and carries direct implications for model selection, risk management, and portfolio construction.

The paper makes three contributions. **First**, I propose a leverage direction taxonomy spanning equities, precious metals, bonds, and cryptocurrency. Gold exhibits a statistically significant inverted leverage effect ($t = -5.79$) that is regime-dependent—inverted during fear-driven rallies but standard during liquidation-driven declines. This taxonomy determines optimal GARCH specification: GJR-GARCH outperforms symmetric GARCH only when $\gamma > 0.10$, correctly classifying all twelve Diebold–Mariano comparisons across two out-of-sample periods for seven primary assets, validated on up to 26 assets.

Second, I link leverage direction to the economic channel of volatility targeting (VT) alpha. Within equity-type assets, γ predicts whether VT acts as trend-following or contrarian (Spearman $\rho = 0.886$, $p = 0.019$), while VT’s primary benefit—maximum drawdown reduction—is universal and driven by base volatility level ($\rho = 0.944$), independent of γ . These results delineate a complexity ceiling: increasing model sophistication improves statistical measurement but not investable allocation decisions.

Third, I document a persistent time-zone arbitrage channel in which U.S. market information spills over to Asian equity markets with a one-day delay, generating t -statistics exceeding the Harvey et al. (2016) multiple-testing threshold in six Asia-Pacific markets.

The manuscript is well suited for the *Journal of Banking and Finance* because the cross-asset risk management analysis—encompassing GARCH model selection, VaR backtesting under Basel III, and volatility targeting—addresses topics central to the journal’s scope, and the practical model selection rule and implementable portfolio strategies provide direct implications for risk managers and institutional investors.

The manuscript has not been submitted elsewhere and is not under consideration at any other journal. I suggest the following potential referees:

1. **Peter R. Hansen** (University of North Carolina at Chapel Hill) — Volatility model comparison and GARCH forecast evaluation.
2. **Alan Moreira** (University of Rochester, Simon Business School) — Volatility-managed portfolios, which my paper extends to cross-asset leverage analysis.
3. **Dirk G. Baur** (University of Western Australia) — Gold safe-haven dynamics and cross-asset volatility.

Thank you for considering my work. I look forward to hearing from you.

Sincerely,

Yi-Hao Lai
Department of Finance, Da-Yeh University
Email: yhlai@mail.dyu.edu.tw